Retinal Screening and Referral Support: Final Product Technical Report

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Repository: github.com/Mikekimm/michael-kimani-eye-screening-capstone (branch `main`) · Video: **[link]**

Scope of claims. This is screening and referral *support*, not a diagnostic device. The model was trained and evaluated only on public datasets; there has been no field deployment or clinical validation. Every figure below comes from a recorded run in the repository.

1. Scope alignment and feature validation

The approved proposal planned an offline-first Android app that screens fundus photos for diabetic retinopathy (DR), glaucoma and cataract on the device, refers uncertain or high-risk cases to a specialist dashboard, and syncs when connectivity returns. Following supervisor feedback (a feasible prototype with one demonstrable workflow; reconcile offline use with sync and authentication; separate research evaluation from development), effort went into one complete, tested DR workflow, while keeping the multi-disease design real: glaucoma and cataract are registered conditions that report "Model not available yet" until a trained model is added, which needs no code change. The full table with evidence paths is in `docs/scope_table.md`.

Planned feature	Intended purpose	Status	Evidence
DR classifier (MobileNetV2, TFLite)	Detect DR from a fundus photo	Completed	`models/dr/`, real-model API tests
Glaucoma and cataract classifiers	Multi-disease screening	Partial	Patient-separated ODIR-5K training notebook ready and smoke-tested; run pending. App shows "not available" until a model is added (no code change)
Five-band cataract grading	Grade cataract severity	Not implemented	ODIR-5K has no severity labels; glaucoma and cataract are binary only rather than showing invented bands
Risk / severity output	Show urgency	Changed	DR: five-band *estimated* band from the binary score (`RiskPolicy`, boundary tests); others binary
Sensitivity ≥ 85%, specificity ≥ 80%	Safe screening trade-off (RQ3)	Not met	Validation fold: sensitivity 0.542, specificity 0.97 at threshold 0.5; hold-out and calibration not completed
On-device inference	Screen without internet	Changed	Server-side LiteRT; offline photos queued and analysed after sync
Android app (React Native)	Mobile client	Changed	Installable PWA; E2E on Android and iPhone emulation
Register patient, capture image, view result	Core health-worker workflow	Completed	E2E journey 1, screenshots
Offline storage and two-way sync	Work with intermittent connectivity	Completed	IndexedDB queue, idempotent push, decision pull; 10 sync tests
Flag and refer uncertain / high-risk cases	Structured referral	Completed	Automatic and manual referral, workflow tests
Specialist review: approve or override	Close the referral loop	Completed	Plus retake / comment; append-only history; E2E journey 2
Admin: users and analytics	Access control, patterns	Completed	Role management, aggregate-only analytics
FastAPI + PostgreSQL backend	Central store and API	Partial	FastAPI + SQLite done; PostgreSQL via `DATABASE_URL` and Docker written, not yet run
Load balancer	Scale to many clinics	Not implemented	Stateless JWT API can scale horizontally
Sync tested under interruptions	Reliability (objective 4)	Completed	Lost-reply, offline-queue and backoff tests
Usability test with health workers	Check usability	Not yet	SUS study planned

Added beyond the proposal: authentication with role-based access (Argon2, JWT refresh rotation, login throttling), an image quality gate, a SHA-256-verified model registry with model cards, eye laterality and model version on every result, an append-only review log and an audit log (the last three answer the supervisor's data-model corrections).

2. Technical functionality

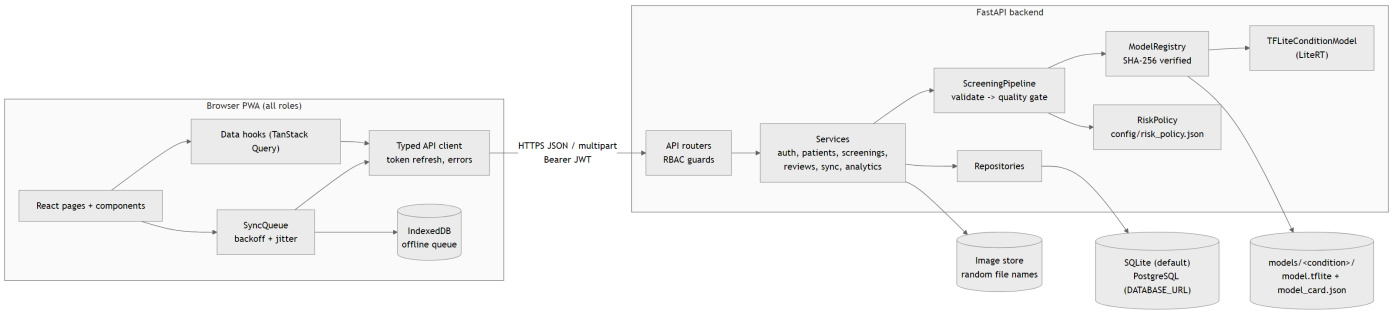


Figure 1. Architecture as implemented (generated from `docs/diagrams/src/architecture.mmd`).

Frontend. A React 18 + TypeScript (strict) PWA, mobile-first (bottom tabs below 1024 px, sidebar above). Pages read server data only through hooks that call one typed API client, which attaches the JWT, refreshes it once on a 401 for all concurrent requests, and converts failures into plain-language messages. Health workers register patients by clinic code, choose the eye, and capture a photo with a circular guide (rear camera or gallery); on-device hints flag dark or blurry photos before upload.

Backend. FastAPI with a layered design: routers (HTTP and role guards) → services (use cases) → repositories (all SQL). `POST /api/screenings` checks type, size and that the bytes really decode as JPEG/PNG, applies the quality gate (blank or tiny images return 422 with a reason), then runs every model the registry has verified and maps each score through that condition's risk policy. One `diagnosis_results` row is stored per condition with its model version, including "not available" rows, so history shows exactly what ran. Positive or borderline results open a referral; specialists approve, override (reason required), request a retake or comment, and each action is appended to `review_events`, which the ORM refuses to update or delete.

Offline and sync. Without a connection, photos and new patients are stored in IndexedDB with a client-generated UUID. A sync engine pushes them in order (patients first), retrying with bounded exponential backoff and full jitter; because the server is idempotent on the UUID, a retry after a lost reply returns the original record rather than a duplicate. Each cycle then pulls specialist decisions made since the last pull. Offline mode captures and queues; it does not run the model on the device.

Data handling and privacy. SQLite by default, PostgreSQL by configuration, schema managed by Alembic migrations. Patients are pseudonymous codes; images are stored under random names and served only to authorised users; analytics and audit records contain no images or patient fields; secrets live only in a git-ignored `.env` generated by a script.

3. Code quality, OOP and algorithms

Principle / pattern	Where
Abstraction + polymorphism	`ConditionModel` → `TFLiteConditionModel`; tests substitute a labelled double
Strategy	`RiskPolicy` per condition, built by `RiskPolicyFactory` from JSON
Open/closed	`ModelRegistry`: new condition = new folder
Repository + DI	backend repositories; `QueueRepository` (IndexedDB / in-memory); `api/deps.py`
Encapsulation	append-only history enforced by ORM listeners; one `AccessPolicy` for record visibility
Single responsibility	routers, services, repositories, presenters, schemas in separate modules

Algorithm	Purpose and complexity
Laplacian variance, brightness, vignette test	quality gate, O(pixels) on a 256^2 copy
Sensitivity-targeted threshold search	highest threshold meeting 85% sensitivity, O(n log n) via sort + binary search
Rank-sum AUC, stratified bootstrap CIs	honest evaluation, NumPy only; AUC matches scikit-learn
Threshold-anchored severity bands	bands can never contradict the decision (1,001-score sweep test)
Idempotent sync, backoff with full jitter	no duplicates, no retry storms
Streaming SHA-256	model integrity, 1 MiB chunks

Quality practices: type hints throughout the backend and TypeScript strict mode with ESLint (which forbids `fetch` outside the API client); Pydantic v2 schemas separate from ORM entities; centralised configuration (settings, risk policy, model cards); small conventional commits on a feature branch (`git log`); a reproducible environment (pinned requirements, `package-lock.json`). Superseded Streamlit code was removed and tagged in git rather than left alongside.

4. Testing approaches and results

Strategy	Why chosen	How	Result
Unit (algorithms)	boundary errors would mis-triage silently	pytest, Vitest; boundary values, property sweep; deliberate code breakage to check tests fail	77 + 25 pass
Integration (API + DB + real model)	security and integrity rules live between layers	fresh DB per test; real LiteRT model; 45-case role-by-route matrix; 9 invalid-upload cases	88 pass
End-to-end functional	prove users can finish clinical tasks	Playwright: worker journey, specialist review, unauthorized access, invalid input	16/16 pass

Compatibility	clinics use varied phones and browsers	same journeys on Chromium (Pixel 5, desktop), Firefox, WebKit (iPhone 13)	all engines pass
Performance	slow hardware and networks	benchmark script; throttled browser round trip; Lighthouse	see section 5
Accessibility	low-vision and screen-reader users	axe-core WCAG 2.1 A/AA on 15 screens, all roles, mobile and desktop	0 violations
Security (brute force)	password guessing	sliding-window login throttle; unit tests with a fake clock and API tests	429 after 5 failures
Model validation	RQ3, honest accuracy claims	validation fold of the training notebook (hold-out notebook prepared, run not finished)	target not met

Different inputs: scores of 0.05, 0.45 and 0.9 (negative, borderline, positive), blank, tiny, corrupt, mislabelled (BMP sent as JPEG), GIF, oversized and future-dated uploads, name-like patient codes, expired, replayed and wrong-type tokens, and every role against every route. **Improvements made because of testing:** a form test exposed that the password field dropped its form reference, so sign-in could never succeed (fixed with `forwardRef`); an end-to-end script test exposed that the calibration script did not write its outputs (fixed); screenshots led to listing DR first and to plain-language referral reasons; Lighthouse raised SEO from 91 to 100; axe-core found unlabeled camera and gallery inputs (fixed); a relative database path that created two databases was anchored.

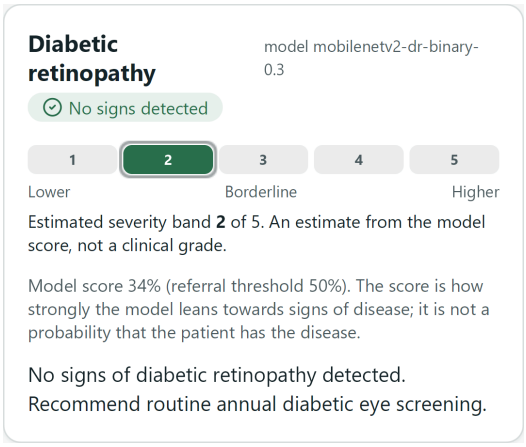


Figure 2. DR result card on iPhone 13 (WebKit), from a synthetic test image: estimated band, model version, score explained, action.

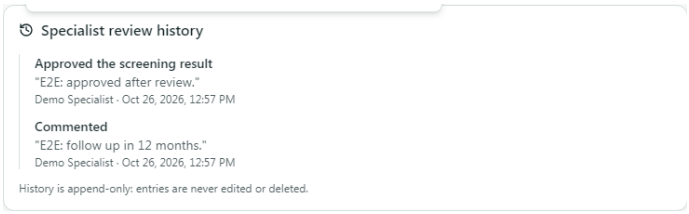


Figure 3. Specialist review history (append-only), desktop Chrome.

5. Performance and compatibility

Measurement (Windows 11, Intel CPU, 8 logical cores)	Result
Model inference, LiteRT, 200 runs (all cores / 1 thread)	55 / 60 ms mean, 68 / 84 ms p95; 2.6 MB model; about 117 MB process memory
Full server pipeline on a 1024 px JPEG (all cores / 1 thread)	74 / 77 ms mean (an earlier run under background load: 166 ms)
Upload to result in the browser (median of 3)	0.50 s; 3.17 s with emulated 6x slower CPU and slow 3G
Lighthouse, sign-in page (performance / accessibility / best practices / SEO)	mobile 96 / 100 / 100 / 100; desktop 100 / 100 / 100 / 100
Accessibility of signed-in screens (axe-core, WCAG 2.1 AA)	0 violations on 15 screens, mobile and desktop
Browsers and devices (Playwright)	Chromium on Pixel 5 and desktop, Firefox desktop, WebKit on iPhone 13: all journeys pass
Bundle	route-level code splitting; core about 110 KB gzipped; charts (103 KB) load only for admins

Analysis: inference is a small share of the round trip, and limiting it to one thread costs about 5 ms, so a modest server CPU is sufficient. On a slow phone and network the wait rises to about 3 s, dominated by the upload; the offline queue means the worker can continue with the next patient meanwhile. Low-end figures are emulated (Chrome DevTools); physical low-end devices and real iOS Safari have not been tested.

6. Model evaluation, limitations and next steps

The DR model is MobileNetV2 (frozen ImageNet backbone, small head) trained for 8 epochs on the Kaggle EyePACS + APTOS + Messidor redistribution with a group-aware 5-fold split. The training notebook reported, on its 47,234-image validation fold at threshold 0.5: accuracy 0.790, precision 0.920, recall 0.542, F1 0.682. These are **validation results from the training notebook, not an independent hold-out**: the same fold guided early stopping, and the independent hold-out run did not finish. Specificity from the same report is 0.97. The **85% sensitivity target was not met** at threshold 0.5: about 46% of DR images in

that fold were missed, which is unacceptable for screening on its own. The prepared remedy is to calibrate the threshold for 85% sensitivity on one half of a verified hold-out and report on the other half, accepting more referrals; this was not completed. The app, model card, About screen and README show these figures with the same label.

- Evaluation is image-level (no patient IDs), the redistribution contains augmented near-duplicates, and the validation fold also guided early stopping, so results are optimistic; there is no external test set and no smartphone-lens images.
- Glaucoma and cataract models depend on the ODIR-5K notebook run (patient-separated split, threshold from validation, test scored once); their results will be reported as measured, even if weak. Offline inference in the browser, PostgreSQL/Docker runs, a usability study with health workers and tests on physical low-end devices remain to be done.
- The TFLite export is quantized and differs from the Keras model by up to 0.034 in score on test inputs; the full-fold difference will be recorded by the hold-out run.